EMWaveNet: Physically Explainable Neural Network Based on Electromagnetic Wave Propagation for SAR Target Recognition

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***Abstract*— Deep learning technologies have significantly improved performance in the field of synthetic aperture radar (SAR) image target recognition compared with traditional methods. However, the inherent lack of interpretability in deep learning models results in low transparency during the decision-making process, which hinders their widespread application in practice. This is especially true in SAR applications, where the credibility and reliability of model predictions are crucial. The complexity and insufficient explainability of deep networks have become a bottleneck for their application. To tackle this issue, this study proposes a physically explainable framework for complex-valued SAR image recognition, designed based on the physical process of microwave propagation. This framework uses complex-valued SAR data to explore the amplitude and phase information and its intrinsic physical properties. The network architecture is fully parameterized, with all learnable parameters endowed with clear physical meanings. Experiments on both the complex-valued MSTAR dataset and a self-built Qilu-1 complex-valued dataset were conducted to validate the effectiveness of framework. The deoverlapping capability of EMWaveNet enables accurate recognition of overlapping target categories, whereas other models are nearly incapable of performing such recognition. Against 0-dB forest background noise, it boasts a 20% accuracy improvement over traditional neural networks. When targets are 60% occluded by noise, it still outperforms other models by 9%. An end-to-end complex-valued SAR automatic target recognition (SAR-ATR) algorithm is constructed to perform recognition tasks in interference SAR scenarios. The results demonstrate that the proposed method possesses a strong physical decision logic, high physical explainability, and robustness, as well as excellent dealiasing capabilities. Finally, a perspective on future applications is provided.**

***Index Terms*— Complex-valued physical explainable deep learning, physical model, synthetic aperture radar automatic target recognition (SAR-ATR).**

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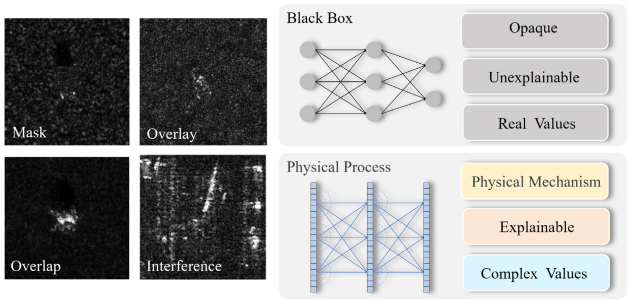
I. Introduction

SYNTHETIC aperture radar (SAR) represents an active microwave sensing technology capable of achieving high-resolution imagery, characterized by its all-weather, all-day imaging. Unaffected by lighting conditions or weather, SAR can provide reliable observational data even under extreme meteorological conditions. Unlike optical remote sensing images, SAR images capture the microwave characteristics of targets, with imaging results influenced by a variety of factors such as polarization modes and wavelengths, which significantly diverge from the forms of images familiar to the human visual system. This divergence leads to considerable challenges in interpretation [1].

In recent years, the rapid development of deep learning technologies has led to increasing application in the field of SAR image interpretation [2]. Through automatic multi-layer feature extraction and learning of specific tasks, deep learning has significantly outperformed traditional SAR image interpretation techniques in terms of performance. However, there exists "hidden layer" between the input data and output results of deep learning models, understanding the mechanisms behind its decisions is challenging. Explainable deep learning models enable us to understand the decision-making process. This transparency significantly reduces decision risks and minimizes the likelihood of misjudgments and errors. By analyzing the model's feedback, potential errors that adversely affect performance can be identified and corrected, thereby optimizing the model's effectiveness and enhancing the credibility and reliability of decision outcomes. Extensive research on explainability has been conducted in fields such as medicine [3] and autonomous driving [4]. In the application of SAR, the lack of explainable target recognition technologies has hindered its practical deployment [5]. Recently, the integration of physical models has been explored in the SAR field to enhance the transparency and reliability of recognition algorithms [6].

This research proposes the design of network architectures using traditional physical models that are highly explainable, the unexplainable feature of "hidden layer" of convolutional neural network (CNN) models. The approach replaces opaque decision-making processes with clearly defined physical processes. To the best of our knowledge, this represents the first attempt to construct a complex-valued physics-based neural

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(a) (b)

Fig. 1. (a) Difficulties in SAR image classification. (b) Traditional CNNs are characterized by opaque internal mechanisms and uninterpretable decision–making processes, while our method is designed for complex-valued SAR images. The parameters learned possess clear physical significance, thereby enhancing the explainability of the network.

network framework for SAR target detection, wherein the network learning process is a strictly physical process and the learned parameters have clear physical meanings as shown in Fig. 1.

The contributions are summarized as follows:

1) A physically explainable neural network architecture is introduced founded on the propagation of electromagnetic waves in passive space for SAR target recognition. The network is fully parameterized with each parameter bearing a clearly physical significance, and the learning and decision-making process of the network is a strictly physical process.
2) A novel loss function is developed named signal-to-noise (SNR) ratio loss L_{SNR} for the network. This function classifies input targets by comparing the energy ratios between designated regions and other regions within the classification layer. All the network parameters are complex-valued and calculated in the frequency domain, effectively using the phase information inherent in SAR images.
3) The network is validated on the MSTAR dataset and self-built Qilu-1 dataset. Experiments were conducted under conditions of target overlap, noise interference, noise overlay, and random masking, with the results being thoroughly analyzed and discussed. This demonstrated an enhancement in the robustness and physical explainability.

The organization of this article is given as follows. The current state of explainable deep learning methods and their application in the field of SAR image interpretation are introduced in Section II. The proposed EMWaveNet is detailed in Section III. The experimental results and discussions are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. Explainable Deep Learning

The explainability is always an Achilles' heel of CNNs and has presented challenges for decades [7]. Selvaraju et al. [8] introduced the gradient-weighted class activation mapping (Grad-CAM), a technique that uses gradient weights to produce class activation maps, highlighting image regions crucial for the model's decision-making. Wang et al. [9] used causal intervention to replace traditional likelihood methods, aiming to eliminate spurious correlations and enhance the model's ability to generalize accurately. Local interpretable model-agnostic explanations (LIMEs) proposed by Mishra et al. [10] construct a local interpretable model near the prediction of interest to approximate the behavior of a complex model. It is model-agnostic, applicable to any model, and provides *post hoc* explanations. The Shapley additive explanations (SHAP) algorithm [11] used game theory to attribute the model's output to individual pixels, revealing each pixel's impact on the overall decision process. Deconvolution is used in neural networks to increase feature map resolution, improving visualization and enabling more precise reconstruction of input images for better explainability [12].

The mathematical essence of the attention model is a weighting strategy for data, which extracts global features compared with the small receptive field area of the CNNs. The attention matrix reflects the characteristics of interest in the decision-making process of the model, making the actions of the model easier to understand. By visualizing attention weight matrices, it becomes possible to pinpoint the specific areas of an image that are relevant to each generated result. The transformer [13], [14], [15], a model entirely constituted by self-attention mechanisms, surpasses CNNs in terms of explainability.

Current AI explainability methods are being applied to enhance model clarity in the SAR field. Visualization techniques (Grad-Cam, LIME, SHAP) link model outputs directly to specific SAR image features to illuminate the decision-making process [16]. In addition, the integration of Bayesian networks [17] for uncertainty quantification and knowledge graphs [18] for structured knowledge representation are used to augment explainability. The adoption of transformers [19], [20] in SAR target recognition improves performance. These methods enhance the interpretability of deep models in the SAR domain, but lack the incorporation of SAR physical information.

B. Physically Explainable Deep Learning for SAR

For SAR imagery, analyses of physical scattering properties can be based on well-founded physical models, such as polarization decomposition models for fully polarized SAR images [21], time–frequency analysis model, and models describing the scattering center properties of targets [22]. These physical models inherently provide explainable feature representations, meaningful priors, and reduce the parameters that networks need to learn [23]. Huang et al. [24] introduced deep SAR net (DSN), a method for SAR image classification that fuses features in both the time and frequency domains. Li and Du [25] developed a method for SAR target recognition that combines global and component information within a deep convolutional neural network framework, enhancing the physical explainability of target features and thus improving model classification accuracy. Zhang et al. [26], Feng et al. [27], [28], and Liu et al. [29] achieved robust performance by integrating scattering information extracted using attribute scattering

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center (ASC) with relevant information from convolutional neural networks. Huang et al. [30] proposed a novel unsupervised approach for infusing physical guidance into deep convolutional neural networks, enhancing the explainability of deep learning models. Chan et al. [31] introduced PCANet, an unsupervised deep convolutional neural network for image classification, where cascaded principal component analysis is used to construct filters. Li et al. [32] applied PCANet to the SAR images, increasing the explainability of model. Liang et al. [33] proposed combining the iterative shrinkage thresholding algorithm with deep convolutional neural networks, enabling the model to automatically learn optimal parameters and thus possess physical explainability. The exploration of uncertainty has been proposed to enhance the explainability of SAR target recognition.

C. Complex-Valued Physically Explainable Framework for SAR

Traditional deep learning models are designed primarily for three-channel RGB optical images and are not well-suited for complex-valued SAR imagery. Many scholars have previously researched how to construct complex-valued deep learning networks to more effectively extract features from complex-valued SAR images [34]. Zhang et al. [35] proposed a complex-valued CNN for SAR imagery, where all the computations are performed in the complex-valued domain and complex-valued label is designed to fit the task, achieving significant results. Wilmanski et al. [36] developed neural networks for SAR target recognition, using a complex-valued first convolution layer and real-valued subsequent layers, with an activation function outputting the magnitude of input feature map. However, to enable the network to learn more physical information, it is essential to incorporate physical models in addition to using complex SAR images. Current research predominantly focuses on the fusion of physical features without embedding physical models into the design of the model framework.

Research on creating an entirely new explainable framework has already been conducted in the optical field. Zhang et al. [37] addressed the issue of convolutional kernel aliasing by grouping kernels, altering the network structure to improve model explainability. They also use decision trees to construct the decision-making process of convolutional neural networks, thereby making the logic of network transparent [38]. Yu et al. [39] introduced MCR2, a method for measuring the size of feature spaces using encoding length, providing a quantitative and mathematically derivable measure for data separability. Building on this method, they derived explainable CNNs [40] and transformers [41], proposing an explainable and stable closed-loop system [42]. The D2NN introduced by Lin et al. [43] is an all-optical network capable of classifying MNIST images in the THz band and can process tasks in parallel [44].

Inspired by these methods and based on the physical imaging mechanism of SAR images, this research designed a complex-valued physically explainable deep learning framework, where all the parameters possess clear physical significance.

III. EMWaveNet

A novel physically explainable deep learning network named EMWaveNet is proposed in this section, leveraging electromagnetic propagation for complex-valued SAR image recognition. The entire framework is illustrated in Fig. 2. EMWaveNet consists of an input layer, an output layer, and several intermediate microwave modulation layers, with detailed descriptions provided in Table I. In addition, Fig. 3 describes the differences between the proposed EMWaveNet and traditional CNN architectures.

The recognition process of the EMWaveNet interprets the complex SAR image as a complex wave field. As the wave propagates through a vacuum, the learnable modulation layers adjust the wave to concentrate its energy within specific regions of the classification layer corresponding to target categories.

A. Theoretical Foundation

In vacuum and source-free conditions, electromagnetic field perturbations under steady-state or frequency-domain analysis can be described by the Helmholtz equation

$$\nabla^2 U + k^2 U = 0 \tag{1}$$

where U represents the electromagnetic field. $\nabla^2 = (\partial^2/\partial x^2) + (\partial^2/\partial y^2) + (\partial^2/\partial z^2)$ is the Laplace operator in 3-D space, a differential operator. k denotes the wavenumber, related to the wavelength and frequency of the wave. The wavenumber is defined as $k = (2\pi/\lambda)$, with λ being the wavelength.

Equation (1) can be solved using integral transform methods, by applying a Fourier transform to $U = U(x, y, z)$ on the xy plane, which correlates the spatial coordinates (x, y) with the frequency domain (f_x, f_y). The result after 2-D fast Fourier transform (2D-FFT) is as follows

$$-4\pi^2\left(f_x^2 + f_y^2\right)G_z + \frac{\mathrm{d}^2 G_z}{\mathrm{d}z^2} + k^2 G_z = 0 \tag{2}$$

where $G_z(f_x, f_y, z)$ represents the 2D-FFT result of $U(x, y, z)$ over the x and y dimensions. Rearranging (2) yields the following expression

$$\frac{\mathrm{d}^2 G_z}{\mathrm{d}z^2} + \left[\frac{2\pi}{\lambda}\sqrt{1 - (\lambda f_x)^2 - (\lambda f_y)^2}\right]^2 G_z = 0. \tag{3}$$

The solution is given by

$$G_z(f_x, f_y, z) = G_z(f_x, f_y, 0)e^{i\frac{2\pi}{\lambda}\sqrt{1-(\lambda f_x)^2-(\lambda f_y)^2}z}. \tag{4}$$

The relationship between the electromagnetic field at various points along the z-axis is established by (4). The electromagnetic field at (x, y, z) can be obtained

$$U(x, y, z) = \mathcal{F}^{-1}\left\{\mathcal{F}[U(x, y, 0)]e^{i\frac{2\pi}{\lambda}\sqrt{1-(\lambda f_x)^2-(\lambda f_y)^2}z}\right\}. \tag{5}$$

The transfer function can be written as

$$H(f_x, f_y) = \exp\left[jkd\left(1 - \frac{\lambda^2}{2}\left(f_x^2 + f_y^2\right)\right)\right] \tag{6}$$

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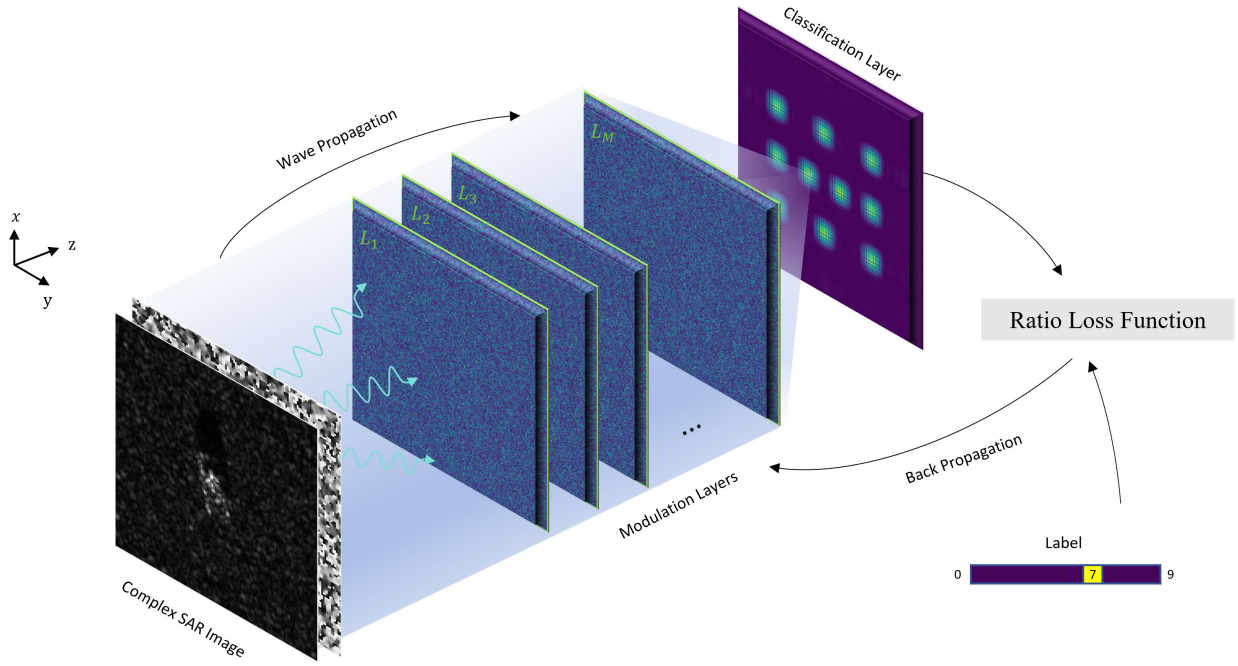


Fig. 2. Overall of EMWaveNet. The network consists of three parts. The input complex-valued SAR image, several modulation layers in the middle and the final classification energy detection layer.

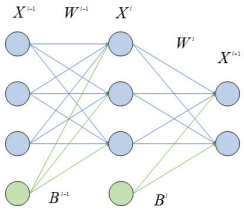
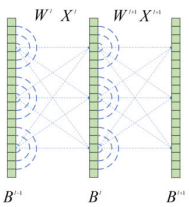
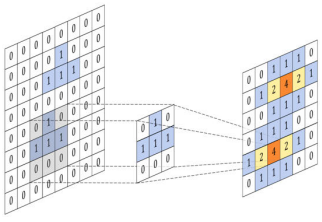
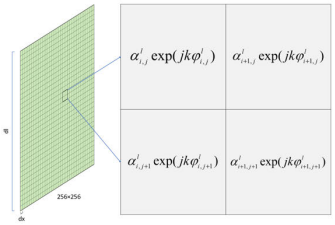
Model	**CNN**	**EMWaveNet**
Theorem	Linear Regression Equation	Electromagnetic Wave Propagation
Formula	$X^{l+1} = F^l(W^l \cdot X^l + B^l)$	$X^{l+1} = W^{l+1} \cdot (X^l \circ B^l)$
Physical Mechanism	-	$U(x,y,z) = \mathcal{F}^{-1}\left\{\mathcal{F}[U(x,y,0)]e^{jkd\left(1-\frac{\lambda^2}{2}(f_x^2+f_y^2)\right)}\right\} \times a(x,y,z)e^{jk\phi(x,y,z)}$
Input	Amplitude SAR Image	Complex-Valued SAR Image
Model		
Layer	Convolution Operation 	Modulate the Amplitude and Phase of Wave 
Loss	$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$	$L_{SNR} = \frac{s^{M+1} - s_C^{M+1}}{s_C^{M+1}}$

Fig. 3. CNN versus EMWaveNet.

where d represents the distance of propagation. Its range is typically set to be on the same order of magnitude as the size of the SAR image to maintain consistency in scale.

Expanding (6) yields

$$H(f_x, f_y) = \exp\left(j\frac{2\pi}{\lambda}z\right)\exp(-j\lambda\pi d\phi_f) \tag{7}$$

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TABLE I
COMPOSITIONAL STRUCTURE OF THE PROPOSED EMWAVENET

Components	Definition	Caption
Input Layer	Complex-Valued SAR image	-
Modulation Layer	Modulate the amplitude and phase of electromagnetic waves	Fig. 5
Classification Layer	Classification is achieved by calculating the regional energy corresponding to each category	Fig. 6

where ϕ_f denotes $(f_x^2 + f_y^2)$. z represents the propagation direction of the wave field. As proven by Sherman [45], the inverse Fourier transform of $H(f_x, f_y)$ is equivalent to the first Rayleigh–Sommerfeld solution

$$h(x, y, z) = \mathcal{F}^{-1}\{H(f_x, f_y)\} = \frac{z - z_i}{r^2}\left(\frac{1}{2\pi r} + \frac{1}{j\lambda}\right)\exp\left(\frac{j2\pi r}{\lambda}\right). \quad (8)$$

The neural network's first convolutional layer is pivotal in learning linear structures and edge features from input images [46]. Notably, in AlexNet, kernels within initial layer exhibit similarities to Gabor filter [47]. Two-dimensional Gabor filter is defined by the product of a plane wave and a Gaussian function

$$G(x, y) = \exp\left(-\frac{x_0^2 + \gamma^2 y_0^2}{2\sigma^2}\right) \times \exp\left(j\frac{2\pi x_0}{\lambda}\right)$$
$$x_0 = x\cos\theta + y\sin\theta$$
$$y_0 = x\sin\theta + y\cos\theta \quad (9)$$

where x and y represent the pixels in the space domain of an image, θ represents the orientation of the stripes, σ^2 is the standard deviation of the Gaussian envelope, and γ represents the spatial aspect ratio.

Inspired by network visualization theory and the Gabor filter, a comparison between (8) and (9) reveals that the propagation formulas for electromagnetic waves in a passive space resemble the Gabor filter. This suggests that the propagation process can also be regarded as a filtering mechanism. This correlation is explicitly discussed in [48]. Thus, the process of the propagation can be regarded as a linear spatial filter capable of extracting specific features from images. This serves as the theoretical foundation for using the propagation of electromagnetic waves in image classification. Combined with the complex-valued characteristics of SAR images and the filtering properties of the transfer function, this study constructs a complex-valued physical neural network based on the transfer function as indicated in (6), which would guide the network integrates the amplitude and phase information into its architecture and processing.

B. Forward Propagation

The forward propagation formula of the traditional neural network can be written as

$$X^{l+1} = F^l(W^l \cdot X^l + B^l) \quad (10)$$

where l indicates the l-th layer within the network, the function F stands for the activation function, W for the weight matrix, X for the input data, and B for the bias.

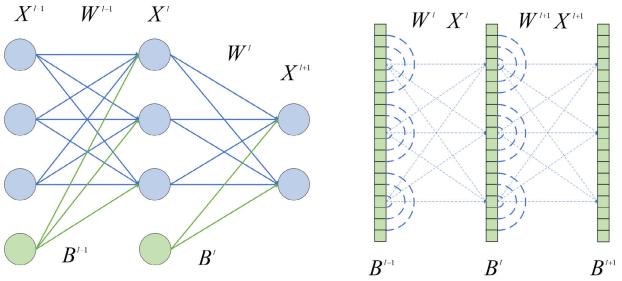


Fig. 4. Comparison of two network structures. (a) Traditional convolutional neural network: The architecture consists of layers of neurons where each neuron in a given layer is connected to the neurons in the previous and subsequent layers. These connections allow the network to learn spatial hierarchies of features through convolution and pooling operations. (b) EMWaveNet: This network is designed to model the propagation of electromagnetic waves in a vacuum, where the process of wave propagation is modulated by the network's parameters. EMWaveNet incorporates the principles of electromagnetic wave theory to enhance the network's ability to capture complex wave behavior in SAR imaging.

Similarly, the forward propagation formula of EMWaveNet can be expressed as

$$X^{l+1} = W^{l+1} \cdot (X^l \circ B^l) \quad (11)$$

where the symbol $\circ$ denotes the Hadamard product, representing elementwise multiplication. A comparison of (10) and (11) reveals that both the networks have two parameters: W and B. However, in CNNs, the bias is added to the weighted sum of data from the preceding layer, whereas in EMWaveNet, the bias is directly multiplied with the input data.

Unlike traditional networks, the weights and biases in the proposed EMWaveNet have clear physical interpretation. The physical expressions are given as follows

$$U(x, y, z) = \mathcal{F}^{-1}\left\{\mathcal{F}[U(x, y, 0)]e^{jkd\left(1-\frac{\lambda^2}{2}(f_x^2+f_y^2)\right)}\right\} \times a(x, y, z)e^{jk\phi(x,y,z)} \quad (12)$$

where $W = \mathcal{F}^{-1}\{\mathcal{F}[U(x, y, 0)]e^{jkd(1-(\lambda^2/2)(f_x^2+f_y^2))}\}$ represents the transmission function of electromagnetic waves in a vacuum. While $B = a(x, y, z)e^{jk\phi(x,y,z)}$, with a and ϕ as the two learnable parameters that modulate the amplitude and phase of the waves. Based on (12), the propagation of microwaves through the various modulation layers of the network as well as the parameter learning process can be simulated.

As illustrated in Fig. 4, the network architectures of the traditional neural network and the proposed EMWaveNet are compared. Although both the networks share the parameters W and B, the traditional neural network learns through convolutional operations, while the proposed network learns layer by layer based on the propagation of electromagnetic waves in space.

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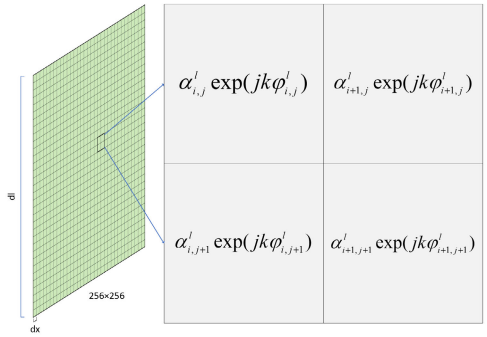


Fig. 5. Detail of the modulation layer. Each neuron in the modulation layer has a set of learnable parameters: amplitude and phase.

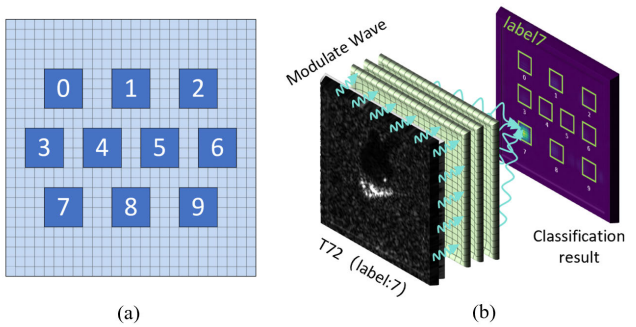


Fig. 6. Classification layer of EMWaveNet. (a) Visualization of classification of the classification layer. Taking the MSTAR dataset as an example, ten regions corresponding to ten target categories are selected on the classification layer. The classification task is realized based on the energy intensity of the microwave signal in the comparison area. (b) Visualization of the classification process on the MSTAR dataset for the T72 target category (label 7) using complex-valued SAR inputs.

Since SAR images are radar images with wave-like properties, a complex-valued SAR image can be regarded as a microwave wavefield. Modulation layers are used to modulate the microwaves originating from the SAR image. Each pixel within the microwave modulation layer can be considered a modulating neuron. These neurons are capable of modulating the amplitude and phase of the wave. The $n_i^l(x, y, z)$ can be used to represent the ith neuron in modulation layer l. It can be calculated by the following

$$n_i^l(x, y, z) = h_i^l(x, y, z) \cdot t_i^l(x_i, y_i, z_i) \cdot \sum_k n_k^{l-1}(x_i, y_i, z_i) \tag{13}$$

where $t_i^l = a_i^l \exp(jk\phi_i^l)$ represents the transmission coefficient, acting as a bias. h denotes the time-domain form of the transfer function.

The forward propagation between layers of EMWaveNet can be represented as

$$n_{i,p}^l = h_{i,p}^l \cdot t_i^l \cdot m_i^l \tag{14}$$

$$m_i^l = \sum_q^l n_{q,i}^{l-1} \tag{15}$$

where n denotes the aforementioned propagation formula for computing the transmission to the next layer. q refers to the qth neuron in modulation layer $l-1$, and p refers to the pth neuron in modulation layer $l+1$. m represents the weighted sum at a certain node in this layer from all the neuronal nodes in the previous layer.

To summarize, the forward propagation process of EMWaveNet is based on the propagation and modulation of the microwave wavefield of a complex-valued SAR image in a vacuum.

C. Modulation Layer

The modulation layer replaces the traditional convolutional layers. While convolutional kernels typically consist of multiple channels, the modulation layer operates with a single channel. Instead of performing convolution operations, the modulation layer adjusts the amplitude and phase of the electromagnetic wave field

$$t(x, y, z) = a(x, y, z)e^{jk\phi(x,y,z)}. \tag{16}$$

The modulation layer is depicted as illustrated in Fig. 5. Each neuron within the modulation layer comprises a set of learnable parameters designed to modulate the amplitude and phase of the electromagnetic waves.

D. Layer for Classification

The output layer is configured as a classification layer to perform classification tasks. With a total of M layers in the network (not including the input and output layers), the output layer is thus the $(M+1)$ layer. The microwave signal received at a neuron on the output layer is represented by m_i^{M+1}, and the energy intensity s_i^{M+1} is given by

$$s_i^{M+1} = \left|m_i^{M+1}\right|^2. \tag{17}$$

As shown in Fig. 6(a), regions are selected on the output layer according to categories c. The input SAR image is classified based on the magnitude of the energy within the regions of the classification layer, as represented in Fig. 6(b).

E. Loss Function

A novel loss function is developed named SNR ratio loss L_{SNR} to replace the cross-entropy loss function of CNNs, tailored for the classification layer of EMWaveNet. The energy for region consistent with the categories c is defined as s_c^{M+1}, and the other energy sum for the layer is represented as $s^{M+1} - s_c^{M+1}$. The loss function is designed as follows

$$L_{\text{SNR}} = \sum_c \frac{\left(s^{M+1} - s_c^{M+1}\right)}{s_c^{M+1}}. \tag{18}$$

Throughout the training process, the value of the loss function continually decreases. This results in an increase in the signal energy value for the corresponding category region, while the values for other regions decrease.

F. Backpropagation

The EMWaveNet is optimized by applying the backpropagation algorithm with the adaptive moment estimation (Adam) optimization method using the loss function. Calculating the gradient of the loss L_{SNR} for both the phase ϕ_i^l and amplitude

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a_i^l of each neuron in layer l. These gradients are then used to iteratively update the phase and amplitude values across the network layers during each training epoch, with the Adam optimizer providing adaptive learning rates for more efficient convergence.

The gradient of L_{SNR} with respect to ϕ_i^l can be written as

$$\frac{\partial L_{\mathrm{SNR}}}{\partial \phi_i^l} = \sum_i \frac{\partial}{\partial \phi_i^l}\left(\frac{s_i^{M+1} - s_i^{M+1} \circ g_i}{s_i^{M+1} \circ g_i}\right) \tag{19}$$

where g_i is a mask associated with the target region (its elements do not depend on ϕ_i^l, and therefore it is constant). Set $N_i = s_i^{M+1} \circ (1 - g_i)$, $D_i = s_i^{M+1} \circ g_i$. Then, (19) can be written as

$$\begin{aligned}\frac{\partial L_{\mathrm{SNR}}}{\partial \phi_i^l} &= \sum_i \frac{D_i \frac{\partial N_i}{\partial \phi_i^l} - N_i \frac{\partial D_i}{\partial \phi_i^l}}{D_i^2} \\ &= \sum_i \frac{D_i\left((1 - g_i) \circ \frac{\partial s_i^{M+1}}{\partial \phi_i^l}\right) - N_i\left(g_i \circ \frac{\partial s_i^{M+1}}{\partial \phi_i^l}\right)}{D_i^2}.\end{aligned} \tag{20}$$

To clearly represent the relationships between layers, we further define i_l for $M+1 \geq l \geq M-L$, where $1 \leq L < M$. L is a parameter that ensures l remains within the total number of microwave modulation layers. Set $m_{i_{M+1}}^{M+1}$ as the weighted sum from all the previous layers at the (i_{M+1})th neuron in the output layer, the derivative of $s_{i_{M+1}}^{M+1}$ with respect to ϕ_i^l can be expressed as

$$\frac{\partial s_{i_{M+1}}^{M+1}}{\partial \phi_i^l} = 2\,\mathrm{Re}\left(\left(m_{i_{M+1}}^{M+1}\right)^* \cdot \frac{\partial m_{i_{M+1}}^{M+1}}{\partial \phi_i^l}\right) \tag{21}$$

where $(\partial m_{i_{M+1}}^{M+1} / \partial \phi_i^l)$ quantifies the sensitivity of the output field with respect to the phase values of neurons in the previous layers.

By applying the chain rule, $(\partial m_{i_{M-L+1}}^{M-L+1} / \partial \phi_{i_{M-L}}^{M-L})$ can be written as

$$\frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial \phi_{i_{M-L}}^{M-L}} = \frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial n_{i_{M-L}}^{M-L}} \cdot \frac{\partial n_{i_{M-L}}^{M-L}}{\partial t_{i_{M-L}}^{M-L}} \cdot \frac{\partial t_{i_{M-L}}^{M-L}}{\partial \phi_{i_{M-L}}^{M-L}}. \tag{22}$$

The gradient can be recursively calculated for each layer, as shown in the following equation

$$\begin{aligned}\frac{\partial m_{i_{M+1}}^{M+1}}{\partial \phi_i^l} &= \frac{\partial m_i^{M+1}}{\partial \phi_{i_M}^M} \cdot \frac{\partial m_{i_M}^M}{\partial m_{i_{M-1}}^{M-1}}, \ldots, \frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial \phi_{i_{M-L}}^{M-L}} \\ &= \sum_{i_M} h_{i_M, i_{M+1}}^M \cdot t_{i_M}^M \cdot \sum_{i_{M-1}} h_{i_{M-1}, i_M}^{M-1} \cdot t_{i_{M-1}}^{M-1}, \ldots, j \\ &\quad \cdot h_{i_{M-L}, i_{M-L+1}}^{M-L} \cdot t_{i_{M-L}}^{M-L} \cdot m_{i_{M-L}}^{M-L}.\end{aligned} \tag{23}$$

Similarly, the gradient of the loss with respect to the amplitude a_i^l of a given layer l is computed as follows

$$\frac{\partial L_{\mathrm{SNR}}}{\partial a_i^l} = \sum_k \frac{2}{g_{i_{M+1}}^{M+1}} \mathrm{Re}\left(m_{i_{M+1}}^{M+1} \cdot \frac{\partial m_{i_{M+1}}^{M+1}}{\partial a_i^l}\right) \tag{24}$$

$$\begin{aligned}\frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial a_{i_{M-L}}^{M-L}} &= \frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial n_{i_{M-L}}^{M-L}} \cdot \frac{\partial n_{i_{M-L}}^{M-L}}{\partial t_{i_{M-L}}^{M-L}} \cdot \frac{\partial t_{i_{M-L}}^{M-L}}{\partial a_{i_{M-L}}^{M-L}} \\ \frac{\partial m_{i_{M+1}}^{M+1}}{\partial a_i^l} &= \frac{\partial m_{i_{M+1}}^{M+1}}{\partial a_{i_M}^M} \cdot \frac{\partial m_{i_M}^M}{\partial m_{i_{M-1}}^{M-1}}, \ldots, \frac{\partial m_{i_{M-L+1}}^{M-L+1}}{\partial a_{i_{M-L}}^{M-L}}\end{aligned} \tag{25}$$

$$\begin{aligned}&= \sum_{i_M} h_{i_M, i_{M+1}}^M \cdot t_{i_M}^M \cdot \sum_{i_{M-1}} h_{i_{M-1}, i_M}^{M-1} \\ &\quad \cdot t_{i_{M-1}}^{M-1}, \ldots, h_{i_{M-L}, i_{M-L+1}}^{M-L} \\ &\quad \cdot \exp\left(j\phi_{i_{M-L}}^{M-L}\right) \cdot m_{i_{M-L}}^{M-L}.\end{aligned} \tag{26}$$

TABLE II
MSTAR DATASET DETAILS

Categories	Series	Depression	Trainset	Testset	Total
BMP2	9563	17°	186	47	233
BTR70	c71	17°	186	47	233
T72	132	17°	186	46	232
T62	A51	17°	239	60	299
BRDM2	E71	17°	238	60	298
BTR60	7532	17°	205	51	256
ZSU23/4	d08	17°	240	59	299
D7	13015	17°	240	59	299
ZIL131	E12	17°	240	59	299
2S1	B01	17°	240	59	299

IV. EXPERIMENTAL RESULTS

A. Dataset and Experimental Settings

In this section, experiments were conducted on the publicly available MSTAR dataset and the self-developed Qilu-1 dataset. Initially, target recognition tasks were performed on both the datasets. Subsequently, the performance of EMWaveNet and other traditional networks was evaluated under conditions of overlap, overlay, masking, and interference. Finally, an end-to-end complex-valued SAR automatic target recognition (SAR-ATR) algorithm was developed, enabling automatic detection and recognition tasks in complex interference scenarios.

1) MSTAR: The MSTAR dataset collected by Sandia National Laboratories SAR sensor platform is selected as the baseline for the study [37]. This SAR sensor operates in the X-band and uses HH polarization and the dataset is widely used in SAR-ATR research. Each data point is a complex-valued image that can be decomposed into amplitude and phase. The image has a resolution of 0.3 × 0.3 m and a size of 128 × 128 pixels, covering 360° in all the directions. The MSTAR dataset contains ten different military vehicle types, namely, rocket launcher: 2S1, armored personnel carrier: BMP2, BRDM2, BTR70, BTR60, bulldozer: D7, tank: T62, T72, truck: ZIL131, and air defense unit: ZSSU234.

In this experiment, images with the depression angle of 17° were selected as the sample dataset, comprising a total of 2747 images. Of these, 2200 images were used for training purposes, while 547 images were designated for testing. For specific details, refer to Table II.

2) Qilu-1: In addition, an SAR aircraft slice dataset was independently developed to corroborate the findings of this research. Data for this dataset were derived from the Qilu-1 satellite, a high-resolution remote sensing satellite autonomously developed by China, equipped with cutting-edge remote sensing imaging technology. It offers high-resolution and high-precision Earth observation data, encompassing an extensive collection of remote sensing images and geographic information data, with broad geographical coverage. A dataset comprising complex-valued SAR image slices of aircraft was constructed using 11 scenes from airports.

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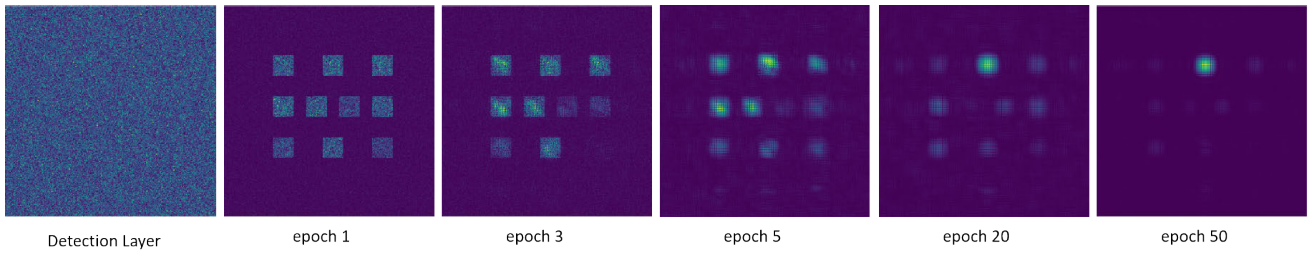


Fig. 7. Visualization of the classification detector layer fitting process. As the number of learning epochs increases, the energy of the microwave signal gradually concentrates in the area to which the input SAR target belongs.

TABLE III
QILU-1 DATASET DETAILS

Categories	Trainset	Testset	Total
A	92	4	96
B	47	4	51
C	57	4	61

TABLE IV
EMWAVENET HYPERPARAMETER SETTINGS FOR DIFFERENT DATASETS

Θ	f	λ	M	N	d	dx	dl
MSTAR	9.6GHz	0.03m	5	256	0.3m	0.0001m	0.256m
Qilu-1	16.7GHz	0.02m	5	256	0.3m	0.0001m	0.256m

In this experiment, 196 images were used for training purposes, while 12 images were designated for testing. The detailed information is referred to Table III.

3) Experimental Settings: All the input images for the experiments were resized to 256 × 256 pixels. The training was optimized using the Adam optimizer with an initial learning rate of 0.1 and conducted over a total of 200 epochs. In addition, our network uses a learning rate decay strategy, wherein the learning rate is periodically adjusted using the StepLR scheduler in PyTorch. Specifically, the learning rate is reduced by a factor of 0.5 every 20 training steps, thereby enhancing the model's performance and stability. All the experiments are conducted on a workstation of the 64-bit Linux operating system with NVIDIA RTX 3090.

B. Target Recognition

1) Hyperparameter Calculation: The hyperparameters required for the initialization of the EMWaveNet network can be calculated based on the radar parameters provided by the dataset. A set of hyperparameters $\Theta = [f, \lambda, M, N, d, dx, dl]$ need to be determined. f is the radar operating frequency, λ is the wavelength, M represents the number of modulation layers, and $N \times N$ is the resolution of the image. d denotes the distance between layers, and dx is the spatial sampling interval, which must be less than half the wavelength according to the Nyquist theorem. dl is the length of the layer, with $dl = N \times dx$. The set of hyperparameters used in the experiments of this article is shown in Table IV.

2) Recognition Results: Experiments were conducted on the MSTAR and Qilu-1 datasets to validate the recognition performance of EMWaveNet. Fig. 7 visualizes the fitting process of the detector, demonstrating that throughout the training phase, with the reduction of the loss function, the signal energy within the detection layer becomes focused in the area corresponding to the respective category of the target. Fig. 8 displays the visualization of classification results for ten target types from the MSTAR dataset on the detector, indicating that inputs from different categories prompt activation in varying regions of the detection layer. As shown in Table V, several neural networks that have shown exceptional performance on classification tasks were selected for comparison with our algorithm. Our model achieves classification accuracy comparable to traditional networks with fewer parameters and reduced computational complexity, requiring fewer floating-point operations (flops) per inference. Notably, the entire classification process and parameters of our model have specific physical significance.

TABLE V
CLASSIFICATION PERFORMANCE COMPARISON OF VARIOUS MODELS ON MSTAR

	Accuracy(%)	Parameters(M)	Flops(G)
AlexNet	99.82	61.20	3.16
Vgg19	99.64	143.7	19.61
ResNet18	99.37	11.69	2.38
ResNet50	99.10	25.56	8.01
ResNext50	99.46	25.04	8.23
ResNext101	98.19	44.55	14.68
DenseNet121	98.92	25.60	4.12
Vision-Transformer	99.64	86.61	17.60
Swin-Transformer	99.82	88.00	15.40
CV-AlexNet	99.64	122.4	6.32
CV-Vgg19	**99.82**	287.34	39.22
CV-ResNet18	99.58	23.38	19.43
CV-ResNeXt50	99.64	51.12	16.02
CV-DenseNet121	99.28	50.08	16.46
Ours	99.58	**1.31**	**0.12**

TABLE VI
ACCURACY OF MODELS WITH DIFFERENT LAYER DEPTH

No. of Layers	1	3	5	10	15
Acc	45.17	98.91	99.34	**99.58**	99.16

3) Model Configuration:

a) Number of modulation layers M: Similar to other deep learning models, the number of modulation layers directly impacts the model's expressive capacity. As the number of layers increases, the network can learn more complex features and patterns. Optimizing the depth of EMWaveNet requires careful consideration of the tradeoff between model complexity, computational efficiency, and the ability to capture and use the intricate details of microwave signal interactions.

As indicated in Table VI, an increase in the number of layers is associated with improved accuracy. However, beyond 15 layers, a decrease in accuracy is observed. This trend is

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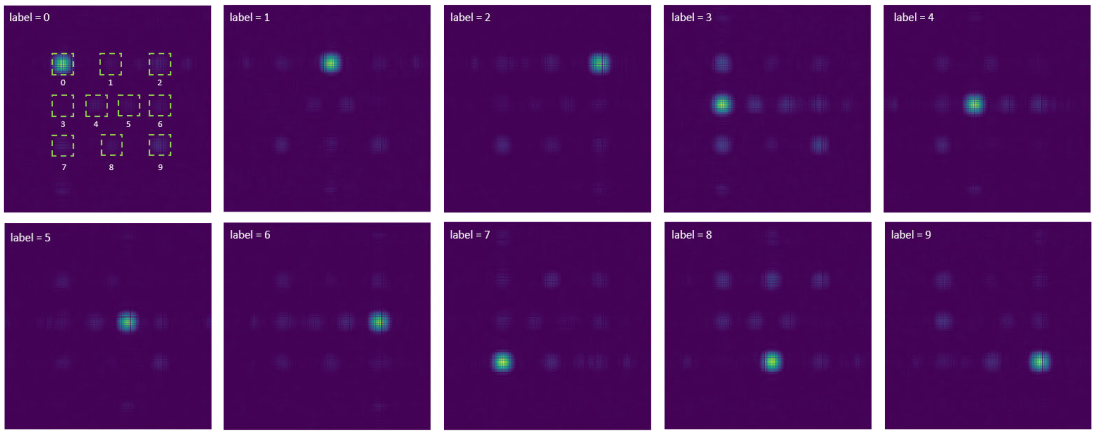
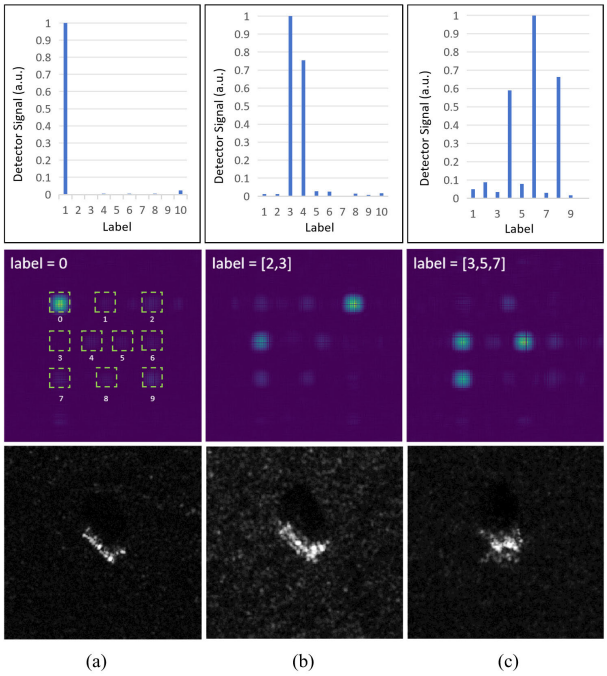


Fig. 8. Visualization results of the classification detection layer for target recognition of different categories in the MSTAR dataset.



(a) (b) (c)

Fig. 9. Visualization of recognition results from the coherent superposition of different complex-valued SAR targets. The first row shows the normalized values of signal energy in the ten regions of the detector. The second row displays the visualization of the classification detection layer, while the third row shows the amplitude graph of the targets' coherent superposition. (a) A single target, labeled as 0. (b) Overlapping of two targets, categories 2 and 3. (c) Blending of three types of targets, categories 3, 5, and 7.

TABLE VII
ACCURACY OF MODELS WITH DIFFERENT MODULATION LAYER SIZES

Mod. Size	64	88	128	192	256	320	512
Acc	98.01	98.73	98.92	99.28	**99.58**	99.10	99.46

attributed to the increasing complexity of network, which leads to overfitting. In the remaining experiments of the article, a depth of five layers was uniformly selected for the network.

b) Modulation layers' size: An increase in the modulation layer size results in more parameters and higher computational cost per layer, thereby enhancing the network's capacity and learning ability.

As shown in Table VII, experiments were conducted with different sizes of the modulation layer. It can be observed that as the modulation layer size increases from 64 to 256, the accuracy gradually improves, with a significant increase from 98.01% to 99.58%. This indicates that increasing the modulation layer size helps the model better learn the features in the data, thereby improving classification accuracy. When the modulation layer size is 256, the model achieves the best accuracy with relatively efficient computational cost. Increasing the modulation layer size further (such as 320 or 512) may lead to increased computational complexity, with limited performance gains, and could even introduce issues such as overfitting. Therefore, in our experiments, choosing a modulation layer size of 256 offers a balance between accuracy and computational efficiency.

TABLE VIII
ACCURACY OF MODELS WITH DIFFERENT COMBINATIONS OF LEARNABLE PARAMETERS

Learnable Parameters	Phi	Amp & Phi
Acc	98.19	**99.58**

c) Learnable parameters: In the modulation layers of EMWaveNet, there are two learnable parameters: $\{a_i^l, \phi_i^l\}$, which modulate the amplitude and phase of the electromagnetic waves, respectively. An experiment was conducted to evaluate the impact of these coefficients on network performance. The results clearly demonstrate that simultaneously learning Amp and Phi significantly improves network effectiveness compared with learning each parameter separately, as shown in Table VIII.

C. Interference Experiment

1) Target Coherent Superposition: Samples from ten categories were randomly selected from the MSTAR trainset and coherently superimposed to create overlapping target images. As illustrated in Fig. 9, the second row of images represents overlapping targets, with "label" denoting their respective categories. It can be observed that the features of the targets are blended together, making them difficult to recognize. The first row presents a visualization of the energy intensity of the classification detection layer of the network. It is evident that the EMWaveNet can easily identify the categories of the constituent targets from the overlapped

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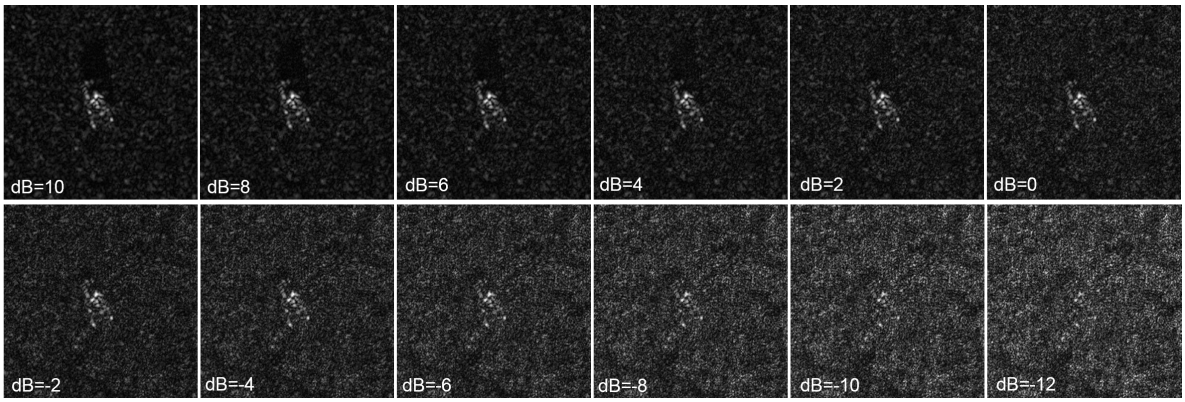
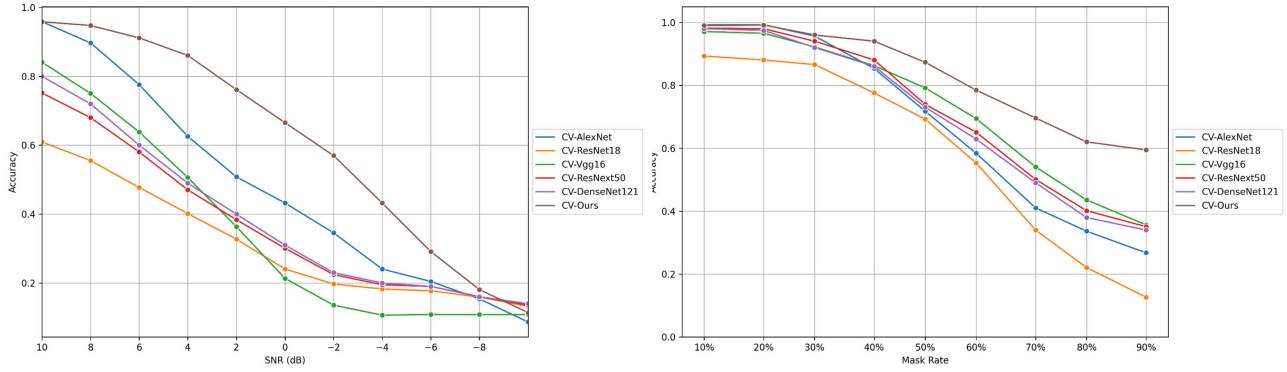


Fig. 10. SAR target images under forest interference with different SNRs.



(a) (b)

Fig. 11. Model accuracy across different forest interferences. Recognition results for (a) forest interference at different SNRs and (b) random-sized interference at a 0-dB SNR.

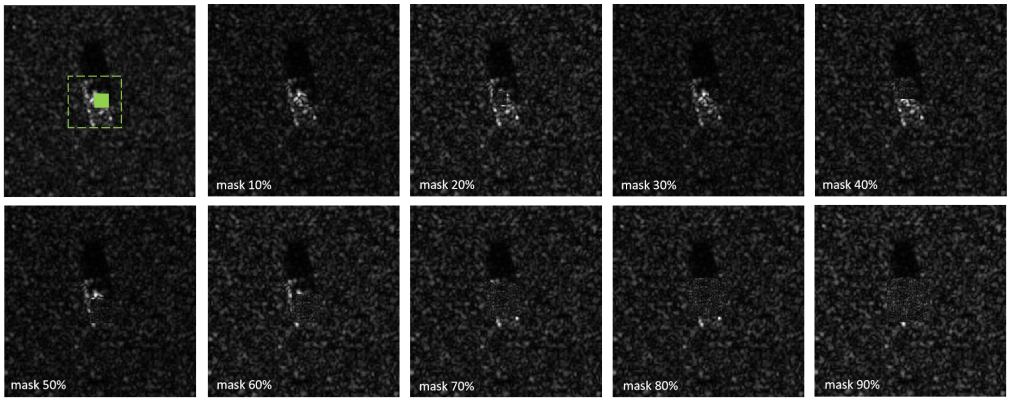


Fig. 12. SAR images with random-sized forest interference at a 0-dB SNR.

images. As shown in Table IX, traditional deep learning networks can almost not handle overlapped target images. In contrast, our method accurately recognizes overlapping categories, demonstrating superior deoverlapping capabilities. By leveraging wave propagation and superposition principles, it surpasses spatial feature-based approaches, effectively distinguishing overlapping targets through energy distribution. This advantage enhances its ability to handle complex interference.

2) Forest Interference: In this study, forest SAR images that align with the spectral band and polarization mode of the MSTAR dataset were used to introduce interference. These images were cropped to match the dimensions of the original targets and then coherently superimposed onto the original complex-valued images, creating interference and simulating the environmental conditions of targets obscured by forest canopy. To assess the impact of forest noise on recognition outcomes, varying levels of forest interference were simulated using different SNRs, defined as follows

$$\mathrm{SNR} = 10\log_{10}\frac{P_{\mathrm{singal}}}{P_{\mathrm{noise}}} = 10\log_{10}\frac{\sum x^2}{\sum n^2}. \tag{27}$$

Fig. 10 illustrates the variation in target images under different SNR conditions of forest interference. It can be observed that as the SNR increases, the proportion of noise escalates, leading to a more extensive coverage of the targets by noise. Fig. 11 and Table X reveal the recognition outcomes under various SNR conditions of forest interference. The

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TABLE IX
OVERLAPPED TARGET RECOGNITION RESULTS OF COHERENT SUPERPOSITION OF DIFFERENT NUMBERS OF TARGETS

Overlap Target Count	1	2	3
AlexNet	100	15.56	1.67
Vgg19	100	2.22	0.83
ResNet18	100	2.22	0.83
ResNext50	100	2.22	0.83
DenseNet121	100	2.22	0.83
Swin-Transformer	100	22.22	3.33
CV-AlexNet	100	31.15	7.50
CV-Vgg19	100	2.22	0.83
CV-ResNet18	100	2.22	0.83
CV-ResNeXt50	100	2.22	0.83
CV-DenseNet121	100	2.22	0.83
Ours	100	**100**	**78.33**

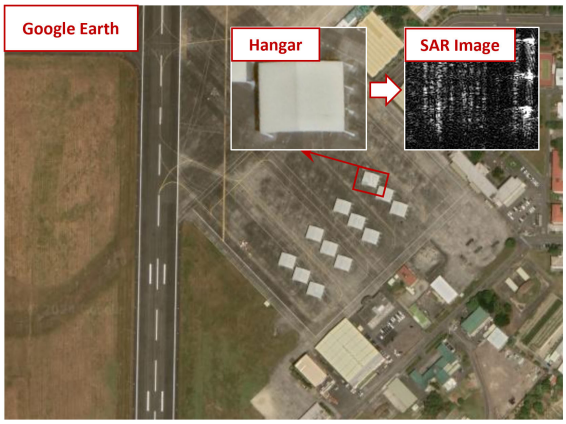


Fig. 13. Optical image of an airfield with some hangars.

performance of EMWaveNet in identifying targets beneath forest canopy has shown a significant improvement compared with traditional neural networks. This further substantiates the enhanced robustness and accuracy of EMWaveNet when tasked with recognizing targets under conditions of noise interference.

3) Random Mask Forest Interference: To simulate more realistic scenarios of targets under the forest, noise characterized by random dimensions is designed to interfere with the targets. The noise interference is constrained to an 88 × 88 area in the center to ensure that all the interferences cover the target. As shown in Fig. 12, forest noise of different sizes is coherently superposed with the target to simulate a more realistic forest semi-interference scene. Fig. 11(b) and Table XI shows the recognition results of different models in the semi-disturbed forest situation. EMWaveNet exhibits better recognition performance as the interference size increases than traditional convolutional neural networks. The greater the interference noise, the more robust the EMWaveNet.

4) Hangar Interference: An experiment involving overlaid hangar interference was designed for the self-built Qilu dataset. As depicted in Fig. 13, this scenario illustrates the situation where aircraft parked within the hangars with their features disrupted and rendered unrecognizable due to interference. Experiments were conducted with the test set aircraft samples overlaid with hangar noise. As shown in Table XII, EMWaveNet is capable of recognizing more aircraft targets disturbed by hangar interference compared with other models. This demonstrates that the algorithm proposed in this study not only possesses high explainability throughout its process but also exhibits strong robustness.

D. SAR-ATR in Complex Interference Scene

The public MSTAR dataset includes not only target chips but also scene images without targets. Both the scene images and target chips are acquired by the same SAR sensor, with a resolution of 0.3 m and HH polarization. Therefore, superimposing the target chips onto the scenes is justifiable. SAR scene data have been simulated with embedded targets in complex interference scenarios for the experiments.

A complex-valued end-to-end SAR-ATR algorithm was designed for the task of target detection and recognition in complex SAR scenes. The algorithm architecture is illustrated in Fig. 14. The input to the network is complex-valued SAR scene images, and the output is the recognition results. The network consists of two main components: a complex-valued detection network and a complex-valued recognition network. The backbone of detection network is a binary classification network based on CV-ResNet18, which performs target and background classification through sliding window traversal of the image. Subsequently, the results of the binary classification are fed into the nonmaximum suppression (NMS) network, which is tasked with selecting the optimal bounding boxes. The recognition network is a complex-valued EMWaveNet, which takes the detection results and processes them to produce the final recognition outcomes.

Detection and recognition results are illustrated in Fig. 15. The results demonstrate that for targets not obscured by forest cover, both the networks accurately identify the target categories. However, for targets obscured by forest noise, the CV-ResNet18 correctly identifies only one target, whereas the EMWaveNet correctly identifies four targets. This underscores the strong noise resistance and robustness of the proposed network in complex interference SAR scenarios.

E. Discussion

In the experiments conducted in this section, EMWaveNet demonstrated superior deblurring performance, capable of accurately identifying targets even under strong interference. The reasons are as follows:

1) Complex-Valued: Current deep learning algorithms are primarily designed for optical real-valued images, hence they are unable to use the phase information of SAR images. The network proposed in this article can directly manipulate complex-valued data within the network architecture. This not only preserves the complete information of the SAR images but also enhances the ability of model to capture the unique characteristics of SAR imagery.

2) Fully Parameterized: Unlike traditional neural networks, the network designed in this study does not use dropout, downsampling, or similar operations but relies on the propagation of microwaves for interaction between layers. Thus, throughout the learning process of the network, no parameters are lost, which benefits the ability of network to recognize superimposed targets.

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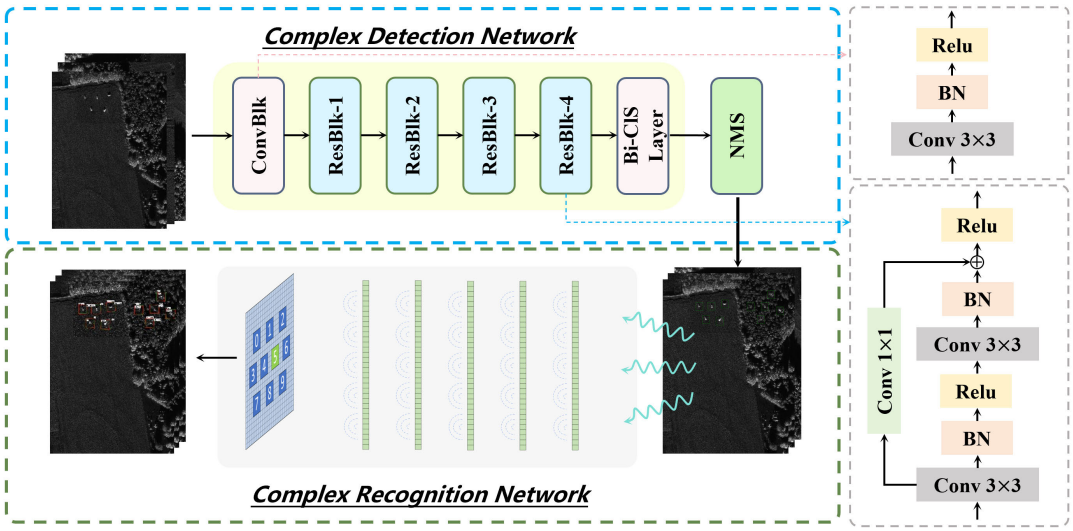
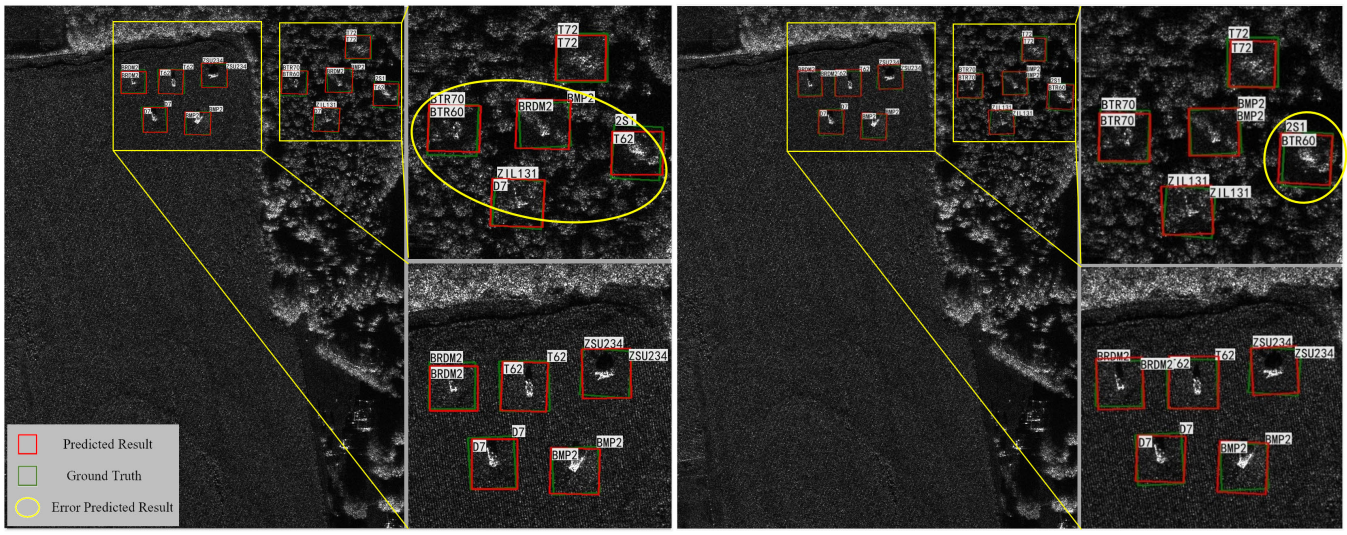


Fig. 14. Overall framework of the end-to-end complex-valued SAR-ATR system. This system is composed of two main parts: a detection network and a recognition network. The input is complex-valued SAR scene images, and the output is the recognition results of the targets.



(a) (b)

Fig. 15. Detection and recognition results in complex interference SAR scenarios. Backbone of the recognition module is (a) CV-ResNet18 and (b) EMWaveNet. In the images, green boxes represent the ground truth, red boxes indicate the detection and recognition results, and yellow circles highlight the erroneous recognitions.

TABLE X
COMPARISON OF CLASSIFICATION PERFORMANCE OF VARIOUS MODELS ON MSTAR UNDER DIFFERENT SNR LEVELS WITH FOREST NOISE OVERLAY

SNR	-10dB	-8dB	-6dB	-4dB	-2dB	-0dB	2dB	4dB	6dB	8dB	10dB
CV-AlexNet	8.68	15.37	20.43	24.05	34.54	43.22	50.81	62.57	77.58	89.69	95.67
CV-Vgg19	10.84	10.84	10.84	10.67	13.56	21.34	36.35	50.63	63.83	75.05	84.09
CV-ResNet18	**13.20**	15.91	17.72	18.26	19.71	24.05	32.73	40.14	47.74	55.52	60.94
CV-ResNext50	13.56	16.04	18.99	19.48	22.44	30.05	38.37	47.04	58.04	67.99	75.17
CV-DenseNet121	14.00	16.56	19.00	20.09	22.99	31.01	40.04	49.15	60.06	72.19	80.22
Ours	11.47	**18.14**	**29.13**	**43.21**	**60.05**	**66.56**	**76.17**	**86.02**	**91.17**	**94.74**	**95.84**

3) Linear Modulation: Electromagnetic wave propagation with dimensional independence facilitates unique linear modulation during spatiotemporal evolution. Consequently, this characteristic enables the EMWaveNet to demonstrate robust parallelism, particularly when overlapping SAR images. This parallelism not only ensures effective separation of overlaid images but also enhances the network's robustness against noise. By increasing the number of layers or neurons, the

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TABLE XI
COMPARISON OF CLASSIFICATION PERFORMANCE OF VARIOUS MODELS ON MSTAR UNDER DIFFERENT RATIO WITH 0-dB FOREST NOISE MASKING

Masking Ratio	10%	20%	30%	40%	50%	60%	70%	80%	90%
CV-AlexNet	**99.28**	**99.28**	95.66	85.35	71.79	58.41	41.05	33.63	26.76
CV-Vgg19	97.11	96.56	92.22	86.26	79.20	69.44	54.07	43.58	35.62
CV-ResNet18	89.33	88.17	86.62	77.58	69.26	55.33	33.99	22.06	12.66
CV-ResNext50	98.20	98.01	94.03	88.07	73.96	65.10	50.10	40.14	35.08
CV-DenseNet121	98.01	97.47	92.04	86.08	73.06	62.93	49.01	37.97	34.00
Ours	99.01	99.19	**96.02**	**94.03**	**87.34**	**78.48**	**69.62**	**62.03**	**59.49**

TABLE XII
RECOGNITION RESULTS OF THE MODELS ON THE QILU-1 DATASET

Model	CV-AlexNet	CV-Vgg19	CV-ResNet18	Ours
No Hangar	100	100	100	100
Hangar	41.6	33.3	33.3	**83.3**

complexity of linear expressions within the network can be further enhanced, thereby improving its modulation capabilities.

V. CONCLUSION

Addressing the issue of current deep learning networks lacking in physical mechanisms and explainability in the SAR domain, this study introduces an explainable framework for SAR image recognition based on microwave propagation named EMWaveNet. The parameters learned within this framework possess distinct physical significance: they modulate the amplitude and phase of microwave propagation. The network inputs are complex-valued SAR images, using the complex-valued information of SAR imagery to uncover latent physical features and enhance the physical logic behind decision-making. The independent propagation characteristics and the superposition properties of electromagnetic waves enable the network to possess deoverlapping capabilities and strong robustness. Its robust performance offers a promising outlook for applications in complex SAR scenarios. Currently, EMWaveNet is presently in the early stages of development. Future research will aim to investigate additional applications of the network's deoverlapping capabilities within the field of SAR.

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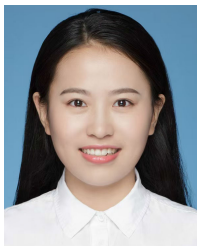
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